No.
DATE / /

# NA-A8 solution

Q1. (a) (i)

~~$P_2(x) = f(x-h)$~~

$P_2(x) = f(x_0) + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$

let $x_0 = x-h$, $x_1 = x$, $x_2 = x+3h$

| | | | |
|---|---|---|---|
| $x-h$ | $f(x-h)$ | | |
| $x$ | $f(x)$ | $\frac{f(x)-f(x-h)}{h}$ | |
| $x+3h$ | $f(x+3h)$ | $\frac{f(x+3h)-f(x)}{3h}$ | $\frac{f(x+3h)-4f(x)+3f(x-h)}{12h^2}$ |

$f'(x) \approx P_2'(x) = f[x_0, x_1] + 2f[x_0, x_1, x_2]\cdot x - f[x_0, x_1, x_2](x_0+x_1)$

$= \frac{f(x)-f(x-h)}{h} + \frac{f(x+3h)-4f(x)+3f(x-h)}{12h^2}(2x-x_0-x_1)$

$= \frac{-3f(x+3h) + 8f(x) - 9f(x-h)}{12h}$

(ii) $f'(x) \approx Af(x-h) + Bf(x) + Cf(x+3h)$

$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(c_1)$

$f(x+3h) = f(x) + 3hf'(x) + \frac{9h^2}{2}f''(x) + \frac{27h^3}{6}f'''(c_2)$

so $f'(x) \approx (A+B+C)f(x) + (-Ah+3Ch)f'(x) + (\frac{h^2}{2}A + \frac{9h^2}{2}C)f''(x)$

$$\begin{cases} A+B+C=0 \\ -Ah+3Ch=1 \\ \frac{h^2}{2}A + \frac{9h^2}{2}C = 0 \end{cases} \Rightarrow \begin{cases} A = \frac{-9}{12h} \\ B = \frac{8}{12h} \\ C = \frac{1}{12h} \end{cases}$$

$f'(x) \approx \frac{f(x+3h) + 8f(x) - 9f(x-h)}{12h}$

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DATE / /

Q1 (b) infinite

$$f(x-h)=f(x)-f'(x)h+\frac{1}{2}f''(c_1)h^2 \quad c_1\in(x-h,x)$$

$$f(x+3h)=f(x)+3f'(x)h+\frac{1}{2}f''(c_2)(3h)^2 \quad c_2\in(x,x+3h)$$

let $f'(x)\approx Af(x-h)+Bf(x)+Cf(x+3h)$

$$\Rightarrow \begin{cases} A+B+C=0 \\ -Ah+3Ch=1 \end{cases} \Rightarrow \begin{cases} A+B+C=0 \\ -A+3C=\frac{1}{h} \end{cases}$$

let $A=\frac{a}{h}$ $\forall a\in R$.

$$\Rightarrow C=\frac{1+a}{3h} \qquad B=-\frac{1+4a}{3h}$$

$$f'(x)-\frac{3af(x-h)+(1+a)f(x)-(1+4a)f(x+3h)}{3h}$$

$$=\left(\frac{3a}{2}f''(c_1)h^2+\cdots-\frac{a}{2}(1+4a)f''(c_2)h^2\right)/3h$$

$$=\frac{a}{2}f''(c_1)h-\frac{3}{2}(1+4a)f''(c_2)h$$

$$=O(h) \quad \leftarrow \text{first order}$$

$\therefore$ let $a=1$

~~$A=\frac{1}{h}$~~

$$f'(x)\approx\frac{3f(x-h)+2f(x)-5f(x+3h)}{3h}.$$

NO

DATE / /

Q2 $f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{1}{6}f'''(c_1)h^3 \quad c_1 \in (x, x+h)$

$f(x-2h) = f(x) - 2f'(x)h + \frac{1}{2}f''(x)(2h)^2 + \frac{1}{6}f'''(c_2)(2h)^3 \quad c_2 \in (x-2h, x)$

so

$$E = f'(x) - \frac{4f(x+h) - 3f(x) - f(x-2h)}{6h}$$

$$= f'(x) - f'(x) + \frac{1}{9}f'''(c_1)h^2 + \frac{2}{9}f'''(c_2)h^2$$

$$= \left(\frac{1}{9}f'''(c_1) + \frac{2}{9}f'''(c_2)\right)h^2$$

$\Rightarrow E = O(h^2)$ order is 2.

Q3. $\int_0^{\pi} x\cos x\,dx$

(a) the composite Midpoint Rule with $m = 1, 2, 3$.

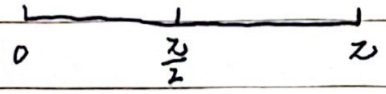
when $m=1$ $h = \pi$. $w_1 = \frac{\pi}{2}$




$\int_0^{\pi} x\cos x\,dx \approx \pi \cdot f(\frac{\pi}{2}) = 0.$

when $m=2$ $h = \frac{\pi}{2}$

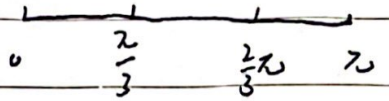
$\Rightarrow w_1 = \frac{\pi}{4}$ $w_2 = \frac{3}{4}\pi.$




$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{2}(f(\frac{\pi}{4}) + f(\frac{3}{4}\pi)) = -\frac{\sqrt{2}}{8}\pi$

when $m=3$ $h = \frac{\pi}{3}$

$\Rightarrow w_1 = \frac{\pi}{6}$ $w_2 = \frac{\pi}{2}$ $w_3 = \frac{5}{6}\pi.$




$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{3}(f(\frac{\pi}{6}) + f(\frac{\pi}{2}) + f(\frac{5}{6}\pi))$

$= -\frac{\sqrt{3}}{9}\pi^2$

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we know that $\int_0^{\pi} x\cos x\,dx = -2$

$m=1$, error $= 2$

$m=2$, error $= \left|-\frac{\pi}{8}\pi^2+2\right|$

$m=3$, error $= \left|-\frac{\pi}{9}\pi^2+2\right|$

(b) the composite Trapezoid Rule with $m=1,2,3$

when $m=1$ $x_0=0$ $x_1=\pi$ $h=\pi$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{2}(f(0)+f(\pi)) = -\frac{\pi^2}{2}$$

error $= \left|-\frac{\pi^2}{2}+2\right|$

when $m=2$ $x_0=0$ $x_1=\frac{\pi}{2}$ $x_2=\pi$ $h=\frac{\pi}{2}$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{4}(f(0)+2f(\tfrac{\pi}{2})+f(\pi)) = -\frac{\pi^2}{4}$$

error $= \left|-\frac{\pi^2}{4}+2\right|$

when $m=3$ $x_0=0$ $x_1=\frac{\pi}{3}$ $x_2=\frac{2}{3}\pi$ $x_3=\pi$ $h=\frac{\pi}{3}$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{6}(f(0)+2f(\tfrac{\pi}{3})+2f(\tfrac{2}{3}\pi)+f(\pi)) = -\frac{2}{9}\pi^2$$

error $= \left|-\frac{2}{9}\pi^2+2\right|$

(c) the composite Simpson's Rule with $m=1,2,3$.

when $m=1$ $x_0=0$ $x_1=\frac{\pi}{2}$ $x_2=\pi$ $h=\frac{\pi}{2}$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{6}(f(0)+f(\pi)+4f(\tfrac{\pi}{2})) = -\frac{\pi^2}{6}$$

error $= \left|-\frac{\pi^2}{6}+2\right|$

when $m=2$ $x_0=0$ $x_1=\frac{\pi}{4}$ $x_2=\frac{\pi}{2}$ $x_3=\frac{3}{4}\pi$ $x_4=\pi$ $h=\frac{\pi}{4}$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{8}(f(0)+f(\pi)+4f(\tfrac{\pi}{4})+4f(\tfrac{3}{4}\pi)+2f(\tfrac{\pi}{2}))$$
$$= -\frac{(\sqrt{2}+1)}{8}\pi^2$$

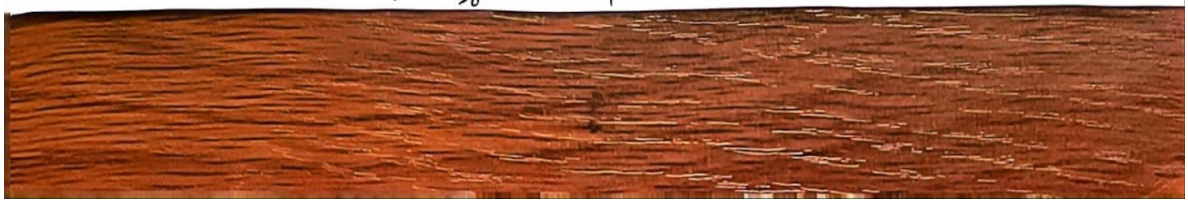
error $= \left|-\frac{(\sqrt{2}+1)}{8}\pi^2+2\right|$

when $m=3$ $x_0=0$ $x_1=\frac{\pi}{6}$ $x_2=\frac{\pi}{3}$ $x_3=\frac{\pi}{2}$ $x_4=\frac{2}{3}\pi$ $x_5=\frac{5}{6}\pi$ $x_6=\pi$

$h=\frac{\pi}{6}$

$$\int_0^{\pi} x\cos x\,dx \approx \frac{\pi}{12}\left(f(0)+f(\pi)+4(f(\tfrac{\pi}{6})+f(\tfrac{\pi}{2})+f(\tfrac{5}{6}\pi))+2(f(\tfrac{\pi}{3})+f(\tfrac{2}{3}\pi))\right)$$
$$= -\frac{(7+\sqrt{3})\pi^2}{36}$$

error $= \left|-\frac{(7+\sqrt{3})\pi^2}{36}+2\right|$



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